

Machine Learning (911.236), Statistical Learning Theory (536.505), Advanced Machine Learning (911.936)

Exercise sheet D

Exercise 1.

4 P.

The **Chebychev inequality** states that for a random variable X with mean μ and variance $\text{Var}(X) = \sigma^2$, the probability that X deviates from its mean by more than $k\sigma$ is at most $\frac{1}{k^2}$, i.e.,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

An equivalent formulation of the Chebychev inequality is

$$P(|X - \mu| \geq k) \leq \frac{\sigma^2}{k^2}.$$

Remember that given a domain $\mathcal{X}$, a label set $\mathcal{Y}$, a hypothesis class $\mathcal{H}$ and a loss function ℓ , we define the generalization error and the empirical error (with training set S of size $m > 0$) as

$$L_{\mathcal{D}}(h) = \mathbb{E}_{(x,y) \sim \mathcal{D}}[\ell(h(x), y)], \quad L_S(h) = \frac{1}{m} \sum_{i=1}^m \ell(h(x_i), y_i).$$

Lets assume $0 \leq \ell(h(x), y) \leq 1$ for all $h \in \mathcal{H}$ and $(x, y) \in \mathcal{X} \times \mathcal{Y}$, and that $m\epsilon^2 \geq 2$. Show that for every fixed $h \in \mathcal{H}$, it holds that

$$\mathbb{P}_S \left(|L_S(h) - L_{\mathcal{D}}(h)| \leq \frac{\epsilon}{2} \right) \geq \frac{1}{2}.$$

You should establish this result by applying the Chebyshev inequality in the right way. Hint: First show an appropriate upper bound on the variance of $L_S(h)$, and then apply the Chebyshev inequality. Please explain **all** your steps in detail.